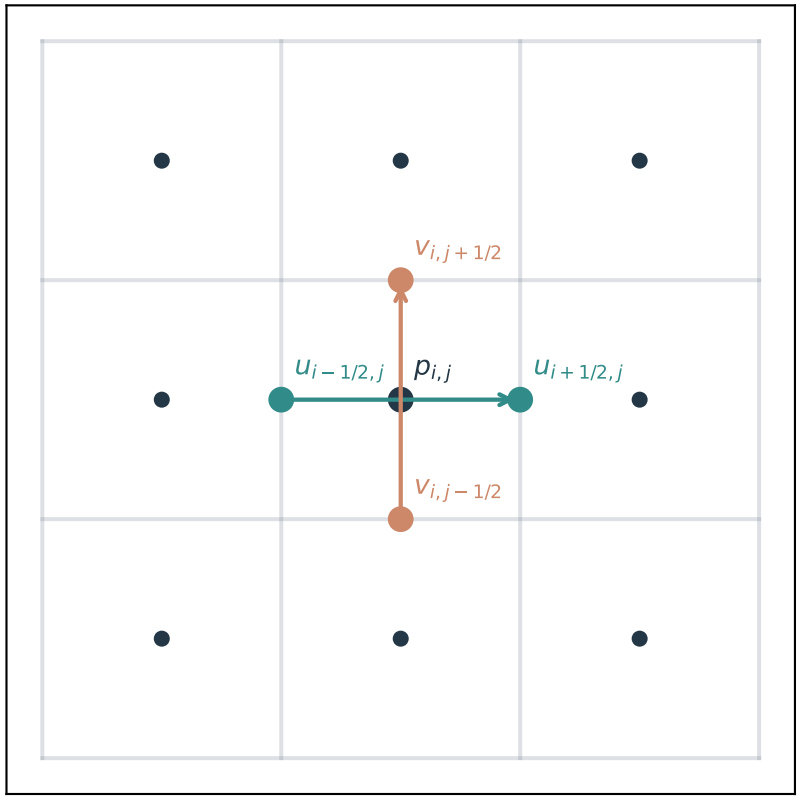
A · Compatible MAC locations



B · Projection geometry and residual

$$G = -D_h^* \quad \Longrightarrow \quad -D_h G \succcurlyeq 0$$

$$P = I - G(D_h G)^{\dagger} D_h$$

$$\|P\mathbf{u}\|_h^2 + \|(I-P)\mathbf{u}\|_h^2 = \|\mathbf{u}\|_h^2$$

$$D_h \mathbf{u}^{n+1} = \Delta t\, r$$

Periodic grid · compatible quadrature · exact pressure solve
With an inexact solve, the final identity quantifies continuity error.